

WAZUH PROJECT DOCUMENTATION STRUCTURE

1. PROJECT OVERVIEW & INFRASTRUCTURE

Virtualized Security Lab Environment :-

Objective: To build a proactive security monitoring system capable of detecting vulnerabilities and automatically mitigating real-time brute force attacks.

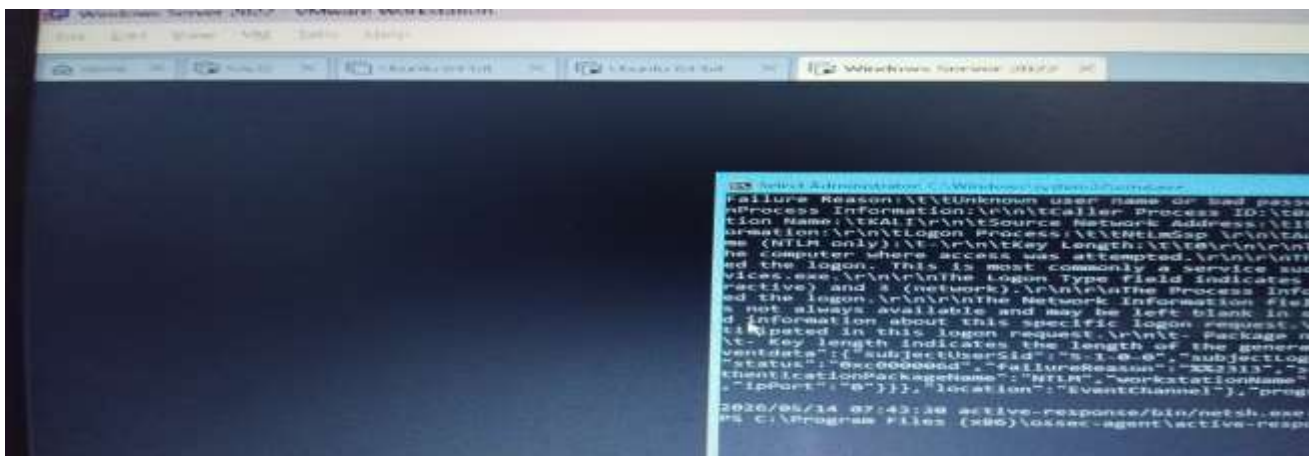
Virtualization Layer: I used VMware Workstation to host an isolated network lab. This ensures that the attacks and configurations remain contained within a sandbox environment, protecting the host machine.

Infrastructure Components :-

Wazuh Manager: Hosted on Ubuntu Server 24.04 LTS. It acts as the central brain for log analysis and decision-making.

Security Agent: Windows Server 2022 (Target Machine). It runs the Wazuh Agent to monitor system logs and file integrity.

Attacker Node: Kali Linux. Used for simulating RDP Brute Force attacks to test the system's resilience.



2. WAZUH MANAGER INSTALLATION

Deploying the Wazuh Central Manager :-

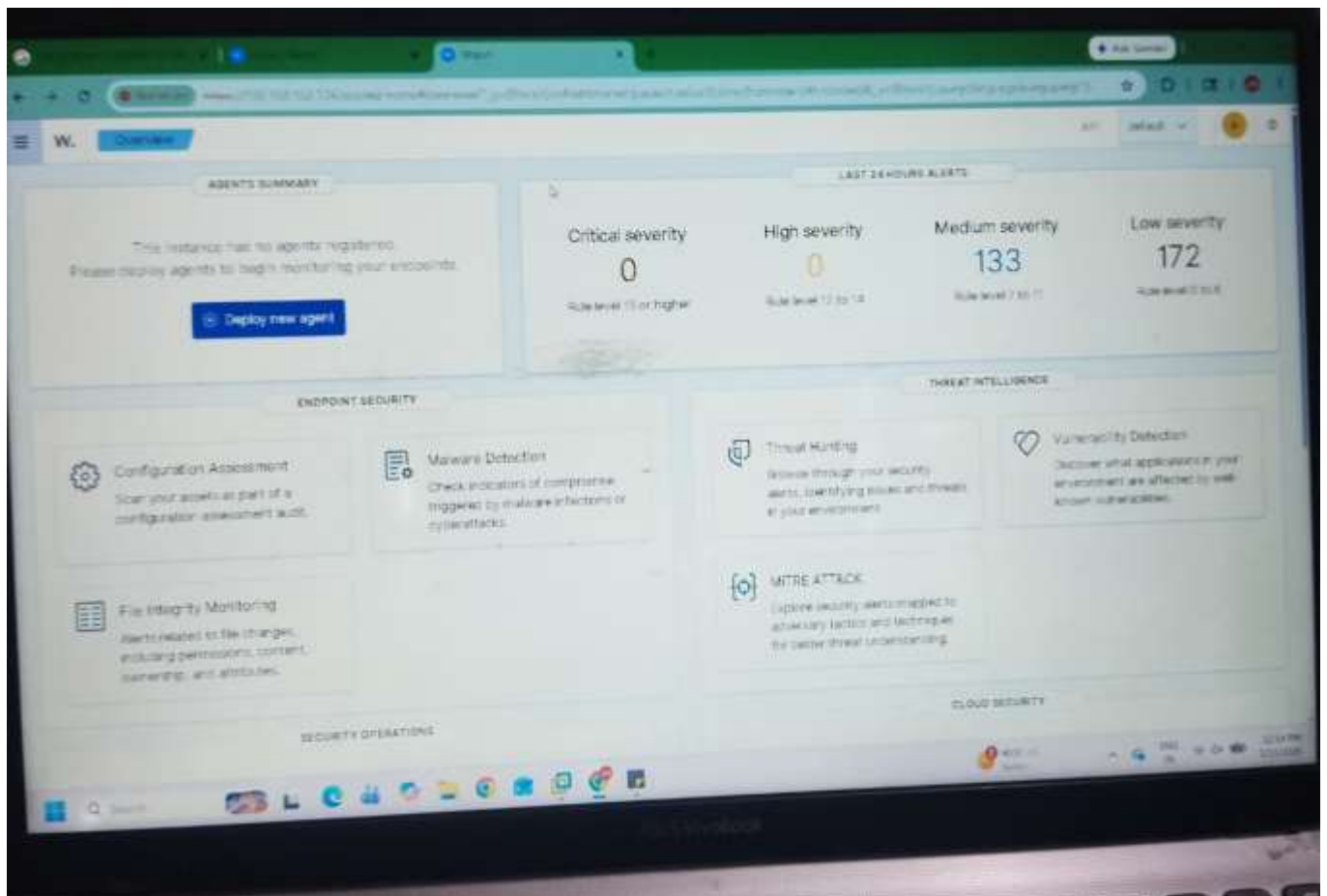
Method: I utilized the Wazuh Installation Assistant for a seamless and error-free deployment.

Configuration: An "All-in-One" installation was performed, which includes the Wazuh Indexer, Wazuh Server, and Wazuh Dashboard on a single Ubuntu instance.

Command Used :

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh  
&& sudo bash ./wazuh-install.sh -a
```

Outcome: The manager was successfully initialized, providing a web-based dashboard for real-time security telemetry.



3. Agent Deployment & Enrollment

Multi-Platform Endpoint Integration :-

Security agents were deployed to monitor endpoints. The enrollment process involved generating unique keys for secure communication between the agents and the manager.

A. Windows Agent Deployment

Method: Deployed via PowerShell using the .msi package.

Command:

```
Invoke-WebRequest -Uri  
https://packages.wazuh.com/4.x/windows/wazuh-agent-  
4.14.5-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i  
$env:tmp\wazuh-agent /q  
WAZUH_MANAGER='192.168.162.134'  
WAZUH_AGENT_NAME='Windows-wazuh'
```

For Start :

```
NET START Wazuh
```

B. Linux Agent Deployment

Method: Deployed via the official Wazuh repository using APT.

Command:

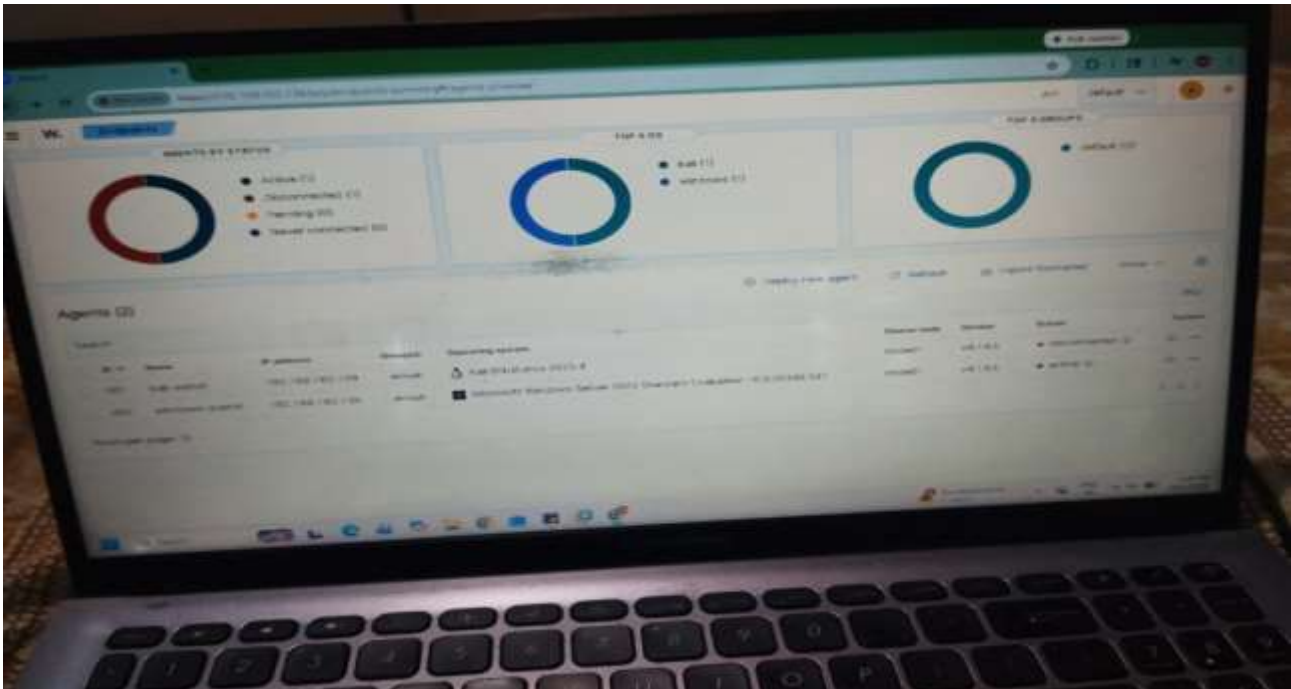
```
wget  
https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-  
agent/wazuh-agent_4.14.5-1_amd64.deb && sudo  
WAZUH_MANAGER='192.168.162.134'  
WAZUH_AGENT_NAME='Windows-wazuh' dpkg -i  
./wazuh-agent_4.14.5-1_amd64.deb
```

For Start :

```
sudo systemctl daemon-reload  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

Status:

Successfully verified that both agents were "Active" on the Wazuh Dashboard.



4. Vulnerability Assessment

Goal:

To find "holes" or weaknesses in the Windows Server before an attacker does.

What we did:

We enabled the Vulnerability Detector module on the Wazuh Manager.

How we did it:

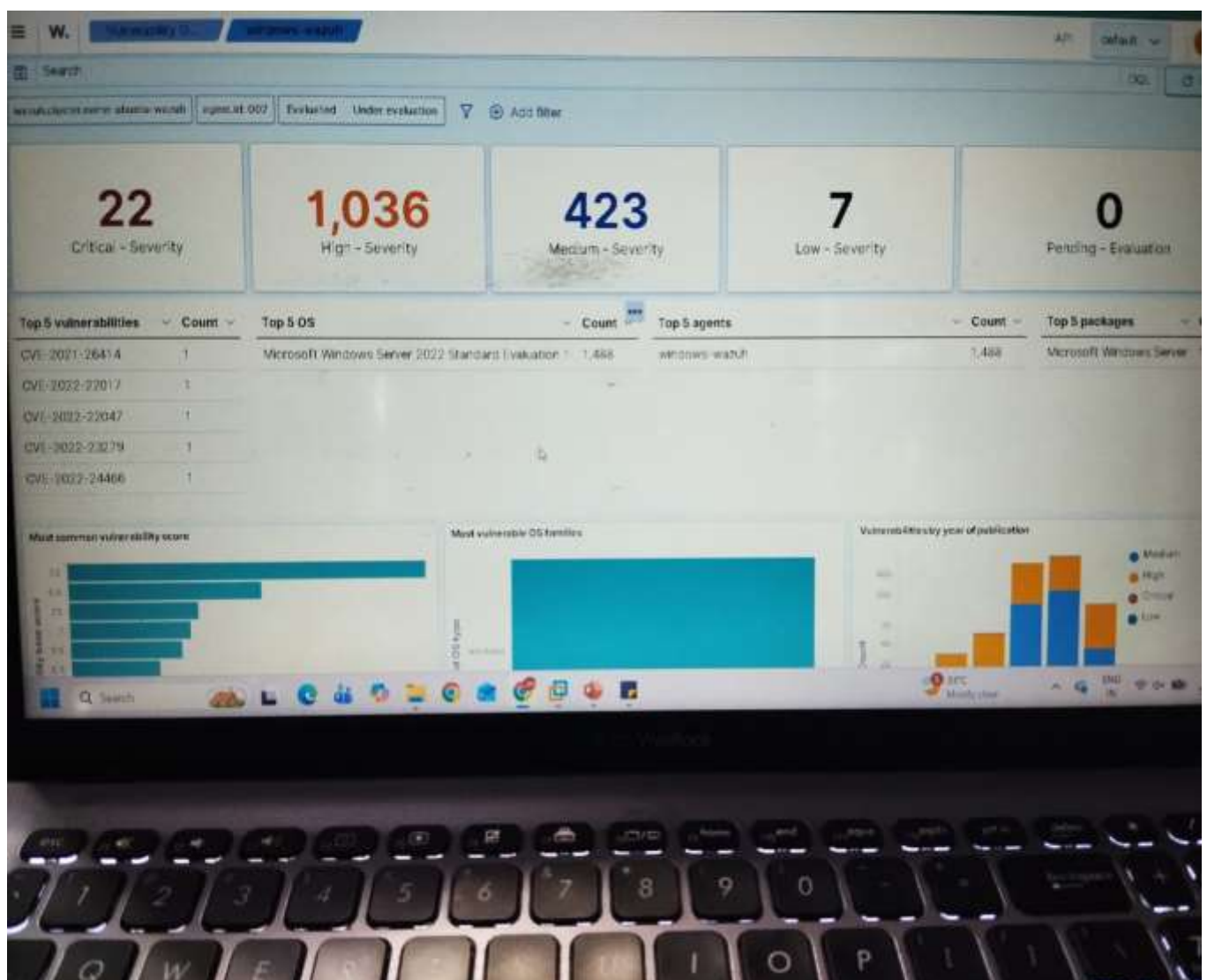
We modified the Manager's configuration to scan the registered agents for known security flaws (CVEs).

The Result:

Wazuh found 22 Critical Vulnerabilities. This proved that the Windows Server was unpatched and highly "hackable," making it an easy target for a Brute Force attack.

Why this matters:

You cannot defend a system if you don't know where it is weak. This step established the "need" for our security system.



5. Active Response Configuration

Goal:

To give the Wazuh Manager the "power" to fight back automatically.

What we did:

We edited the configuration file on the Ubuntu Manager.

File Path: `/var/ossec/etc/ossec.conf`

The Logic:-

Command:

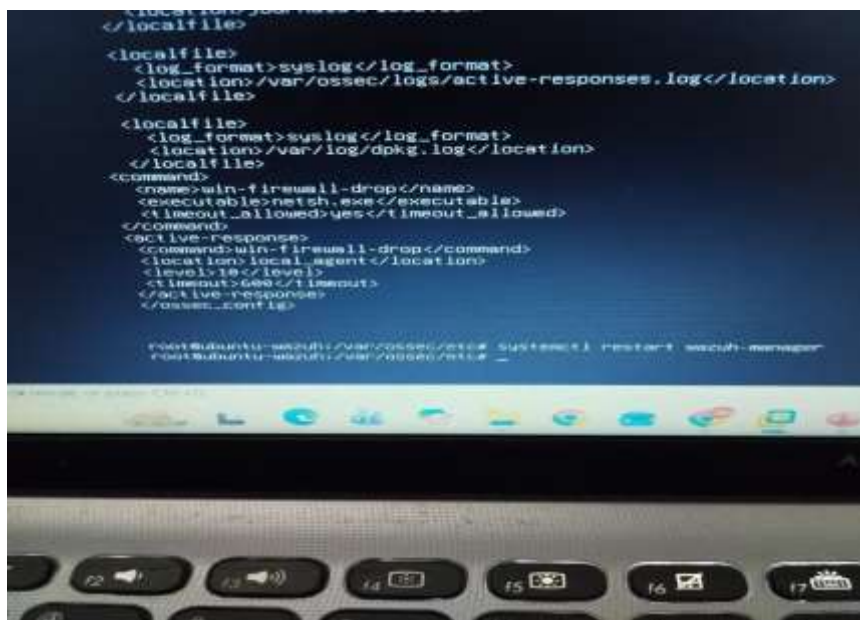
We defined **win-firewall-drop** which uses **netsh.exe** (the Windows firewall tool).

Trigger:

We set the Alert Level to 10. This means the system won't block someone for one wrong password, but it will block them if they try 10 times (Brute Force pattern).

Location:

We set it to `local_agent` so the block happens right on the victim's machine.



Command used to apply changes:

```
systemctl restart wazuh-manager
```

6. Attack Simulation & Automated Mitigation

Goal:

To test if our "Bodyguard" (Wazuh) can actually stop a real attack.

The Attack (Step 1):

We used Hydra on Kali Linux to launch an RDP Brute Force attack.

Command:

```
hydra -l Administrator -P pass.txt rdp://192.168.162.135 -t 2 -vV
```

The Detection (Step 2):

Wazuh Manager saw the flood of failed logins and instantly triggered **Rule 60204 (Level 10)**.

The Action (Step 3):

The Manager sent an "Active Response" command to the Windows Agent. The Agent ran **netsh.exe** and blocked the Kali IP.

The Final Result:

Hydra stopped working and showed "Connection Refused."
The attacker was kicked out automatically in seconds.

